UAV Health Monitoring Task List

Ally Blair (ME) – *Mechanical Systems, Data Collection*

1. **Assemble UAV and test platform** with stable mounting for motors and sensors.
2. **Conduct vibration isolation experiments** to reduce structural noise in audio recordings.
3. **Compare microphone types (MEMS vs USB)** under identical test conditions.
4. **Determine optimal microphone placement** through controlled bench testing.
5. **Induce controlled propeller faults** (e.g., chipped, unbalanced, bent blades) for dataset generation.
6. **Induce controlled motor faults** (e.g., bearing wear, partial short) and record signatures.
7. **Establish safety protocols** for running UAVs with faulty motors during testing.
8. **Record baseline healthy motor datasets** across different operating conditions (RPM, load, environment).
9. **Record faulty motor datasets** at varying fault severity levels for training/evaluation.
10. **Document mechanical test setup** including fault induction procedures, mic placement diagrams, and UAV configuration.

Prissha Chawla (CS) – *Data Science, AI, ML*

11. **Design of experiment plan** detailing recording scenarios, environmental variations, and repetitions.
12. **Perform exploratory data analysis (EDA)** on collected motor audio datasets.
13. **Convert audio data into spectrograms** and evaluate preprocessing pipelines.
14. **Develop semi-supervised autoencoder model** for anomaly detection on healthy motor audio.
15. **Evaluate reconstruction error distributions** to establish anomaly detection thresholds.
16. **Investigate supervised ML approaches** (e.g., DeepONet or alternatives) for UAV battery health monitoring.
17. **Train and validate ML models** using induced fault datasets and battery profiles.
18. **Conduct hyperparameter tuning** for both autoencoder and supervised models.

19. **Benchmark model performance** (precision, recall, F1) against UAV health monitoring objectives.
20. **Document AI methodology and results** for inclusion in project reports and publications.

Siddharth Urankar (CS) – ***Software, Integration, Deployment***

21. **Install and configure Linux OS** on NVIDIA Jetson for UAV onboard computing.
22. **Develop microphone interface software** to support synchronized acoustic data capture.
23. **Implement real-time data logging system** for motor audio during UAV operation.
24. **Optimize data storage format** for efficiency in transfer and later ML model use.
25. **Deploy trained autoencoder model** onto Jetson for onboard inference.
26. **Develop anomaly detection pipeline** that integrates audio capture with autoencoder output.
27. **Test latency and throughput** of model inference on Jetson under flight-like conditions.
28. **Implement battery data collection interface** for supervised RUL prediction model.
29. **Optimize Jetson codebase** to balance recording, inference, and data transfer simultaneously.
30. **Validate end-to-end system integration** by running real-time UAV health monitoring tests.